



Business Insights Report

U.S. AIRLINE PERFORMANCE & DELAY ANALYSIS

Executive Summary

Transforming Data Into Operational Excellence

This project analyzed over **20,000 flight records** to evaluate airline performance, airport efficiency, route behavior, and delay trends.

Using **SQL for data preparation** and **Power BI for visualization**, the analysis identified operational bottlenecks affecting flight punctuality and passenger experience.



Core Objective

Understand the primary factors influencing flight delays and provide actionable recommendations to improve operational efficiency, reduce avoidable delays, and enhance customer satisfaction across the entire network transit chain.

Analytics Workflow Architecture



Data Ingestion

Importing CSV datasets into SQL Server, verifying row counts and structural integrity across 3 primary tables.



ETL & Cleaning

Converting temporal fields, handling null values, and feature engineering to calculate delay duration metrics.



BI Visualization

Connecting Power BI to SQL for real-time KPI tracking and interactive geographical airport performance views.

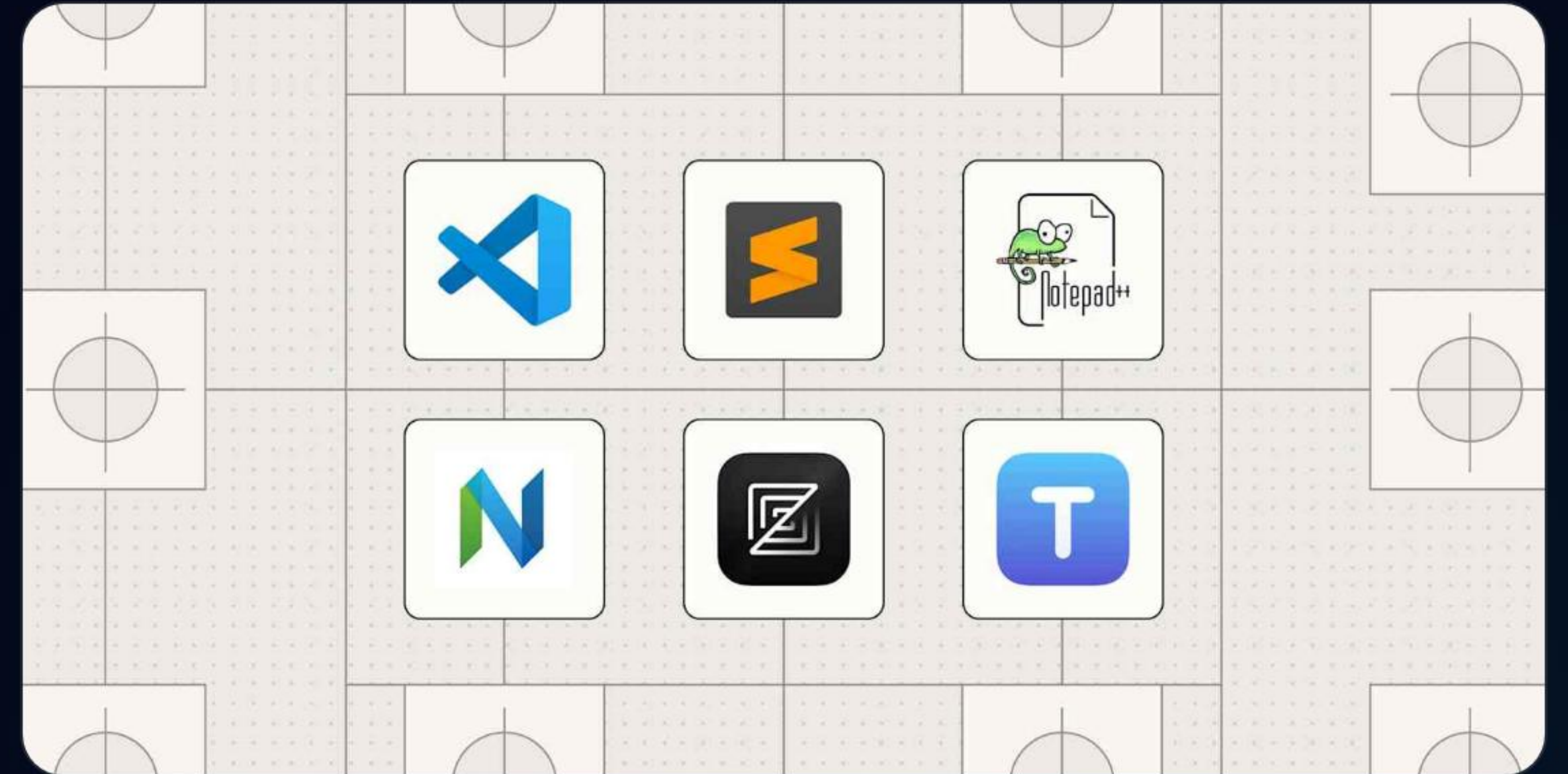
SQL-Driven Data Engineering

Modernizing Flight Data

To enable monthly and seasonal trend analysis, raw year/month/day columns were concatenated and cast into proper date formats.

SQL logic used for structural alignment:

```
CONCAT(YEAR, '-', MONTH, '-', DAY) AS DATE
```



Flight Performance Overview

20K

TOTAL FLIGHTS

8.69

AVG DEPARTURE DELAY

4.90

AVG ARRIVAL DELAY

Average metrics measured in minutes. Study baseline encompasses nationwide network records.

1. Carrier Performance Variance



The Finding

Noticeable differences in average departure delays among airlines. Some carriers maintained stable operations, while others experienced substantially higher spikes.

Business Impact

Frequent delays hurt customer satisfaction, increase compensation costs, and reduce downstream aircraft utilization.

2. Airport Congestion Hotspots

The Finding

A small group of concentrated hub airports accounted for a disproportionately large share of delayed flights, driven by high volume limits.

Business Impact

Airport congestion creates cascading terminal delays across the network, damaging tightly budgeted connecting flights.

3. Temporal Delay Distributions



The Finding

Delays were not distributed evenly over the year. Specific months and peak days experienced noticeably higher operational frictions.

Business Impact

Peak holiday/seasonal travel periods spike workload pressure, multiplying baseline vulnerability risks.

4. Route-Level Critical Bottlenecks

Origin	Destination	Avg Delay (min)	Volume
ABE	DTW	11.00	12
ABQ	ATL	18.33	145
ABQ	DFW	28.33	210
ATL	ABE	52.00	10

The Finding

Several transit route segments consistently recorded higher average delay profiles compared to the rest of the operational network baseline.

Business Impact

Repeated route failures directly trigger localized passenger dissatisfaction and break connecting flight promises.

5. Concentrated Network Vulnerability



The Pareto Distribution Principle

Delays are heavily isolated within specific carriers, airports, and routes rather than being spread globally across the network map.

Business Benefit

Targeted adjustments will generate substantial performance recoveries without requiring sweeping, expensive system-wide process overhauls.

| Strategic Action Items (1 – 3)

-  **Improve Airline Scheduling Efficiency:** Leverage historical delay data trends to optimize aircraft turnaround buffer configurations and crew duty assignments.
-  **Strengthen Airport Resource Planning:** Dynamically allocate gate staffing and ground support infrastructure assets based on forecasted volume peaks.
-  **Monitor High-Risk Routes:** Establish proactive early-warning metric parameters for underperforming routes to modify block times early.

| Strategic Action Items (4 – 5)

-  **Develop Predictive Delay Monitoring:** Use predictive models trained on historical flight behaviors to mitigate potential network disruptions.
-  **Enhance Passenger Communication:** Deliver clear, automated, real-time messaging alerts during flight interruptions to preserve brand trust.

Business Value Delivered



Operational Highlights

- Identified primary friction constraints.
- Isolated core airport delay centers.
- Exposed high-variance flight schedules.



Technical Achievements

- Data-driven decision architecture via BI.
- Built targeted tactical playbooks.
- Transformed messy tracking files into clean BI assets.

Conclusion

By prioritizing tactical efforts on the most impactful operational bottlenecks, operators can structurally reduce systemic delays and elevate passenger service metrics. This project proves the immense worth of combining structured **SQL data engineering** with dynamic **Power BI dashboards** to guide long-term business strategy.

DATA INSIGHTS TO STRATEGIC GROWTH

| Image Sources



Thumbnail

<https://images.ctfassets.net/lzny33ho1g45/1nBwzHo2cVkaJDcuwPhaeY/1a62e16f78007e3cbc60765bc4c8cfbd/best-code-editor-hero.jpg?fm=jpg&q=31&fit=thumb&w=1520&h=760>

Source: zapier.com

for



Thumbnail

https://images.stockcake.com/public/6/f/3/6f3e2b68-c575-466a-87f8-fce49c6ce238_large/global-airport-terminal-stockcake.jpg

Source: stockcake.com

for

stockcake.com